

|                                                                                   |                                                                                                                                                                                    |                                                                                     |
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|  | <div style="text-align: center;"><b>FATE</b><br/><b>MONTHLY REPORT</b></div> <hr/> <div> <div>Doc.No : FATE-001</div> <div>Version : 1.0</div> <div>Date : 08-06-2024</div> </div> |  |
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# FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

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Performance assessment  
Annex  
May 2024



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| wind speed<br>$ms^{-1}$                                                         | Observations   |                |          |
|---------------------------------------------------------------------------------|----------------|----------------|----------|
|                                                                                 | ws < 4.2       | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 3740           | 1373     |
|                                                                                 | 4.2 < ws < 7.6 | 2126           | 2654     |
|                                                                                 | ws > 7.6       | 224            | 1219     |
| Number of points 17447 ; PC=65.1%; EBD=1.9%; POD1=61.4%; POD2=50.6%; POD3=81.1% |                |                |          |

Table 1: night1 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

| wind speed<br>$ms^{-1}$                                                         | Observations   |                |          |
|---------------------------------------------------------------------------------|----------------|----------------|----------|
|                                                                                 | ws < 4.2       | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 5672           | 406      |
|                                                                                 | 4.2 < ws < 7.6 | 418            | 4556     |
|                                                                                 | ws > 7.6       | 0              | 284      |
| Number of points 17447 ; PC=92.2%; EBD=0.0%; POD1=93.1%; POD2=86.8%; POD3=95.9% |                |                |          |

Table 2: night1 - Contingency table for wind speed ( $ms^{-1}$ ) processed with AR (1H)

| wind speed<br>$ms^{-1}$                                                         | Observations     |                  |           |
|---------------------------------------------------------------------------------|------------------|------------------|-----------|
|                                                                                 | ws < 12.0        | 12.0 < ws < 18.0 | ws > 18.0 |
| Model data                                                                      | ws < 12.0        | 15304            | 159       |
|                                                                                 | 12.0 < ws < 18.0 | 90               | 1569      |
|                                                                                 | ws > 18.0        | 0                | 26        |
| Number of points 17447 ; PC=98.0%; EBD=0.0%; POD1=99.4%; POD2=89.5%; POD3=77.3% |                  |                  |           |

Table 3: night1 - Contingency table for wind speed ( $ms^{-1}$ ) processed with AR (1H) and thresholds (12.0,18.0)**0.1 NIGHT1: Contingency tables**

Tables in this section are included in the range: 1-22.

| wind direction<br>degree |       | North | East | South | Observations | West |
|--------------------------|-------|-------|------|-------|--------------|------|
| Model data               | North | 6797  | 421  | 45    |              | 96   |
|                          | East  | 526   | 831  | 438   |              | 5    |
|                          | South | 62    | 289  | 3189  |              | 266  |
|                          | West  | 212   | 60   | 83    |              | 235  |

Number of points 13555 ; PC=81.5%; EBD=1.3%; POD1=89.5%; POD2=51.9%; POD3=84.9%; POD4=39.0%

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

| wind direction<br>degree |       | North | East | South | Observations | West |
|--------------------------|-------|-------|------|-------|--------------|------|
| Model data               | North | 7494  | 131  | 0     |              | 19   |
|                          | East  | 93    | 1394 | 50    |              | 0    |
|                          | South | 0     | 76   | 3699  |              | 28   |
|                          | West  | 10    | 0    | 6     |              | 555  |

Number of points 13555 ; PC=97.0%; EBD=0.0%; POD1=98.6%; POD2=87.1%; POD3=98.5%; POD4=92.2%

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

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| relative humidity                                                               |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| percent                                                                         |                 | rh < 7.1     | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                                                                      | rh < 7.1        | 2182         | 728             | 29        |
|                                                                                 | 7.1 < rh < 15.3 | 2423         | 3940            | 949       |
|                                                                                 | rh > 15.3       | 495          | 1712            | 4891      |
| Number of points 17349 ; PC=63.5%; EBD=3.0%; POD1=42.8%; POD2=61.8%; POD3=83.3% |                 |              |                 |           |

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

| relative humidity<br>percent                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | rh < 7.1     | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                                                                      | rh < 7.1        | 4664         | 249             | 3         |
|                                                                                 | 7.1 < rh < 15.3 | 435          | 5943            | 186       |
|                                                                                 | rh > 15.3       | 1            | 188             | 5680      |
| Number of points 17349 ; PC=93.9%; EBD=0.0%; POD1=91.5%; POD2=93.2%; POD3=96.8% |                 |              |                 |           |

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

| relative humidity                                                               |                  | Observations |                  |           |
|---------------------------------------------------------------------------------|------------------|--------------|------------------|-----------|
| percent                                                                         |                  | rh < 50.0    | 50.0 < rh < 70.0 | rh > 70.0 |
| Model data                                                                      | rh < 50.0        | 16801        | 38               | 0         |
|                                                                                 | 50.0 < rh < 70.0 | 28           | 323              | 4         |
|                                                                                 | rh > 70.0        | 0            | 7                | 148       |
| Number of points 17349 ; PC=99.6%; EBD=0.0%; POD1=99.8%; POD2=87.8%; POD3=97.4% |                  |              |                  |           |

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

| precipitable water vapor<br>mm                                                  |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | pwv < 1.9    | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                                                                      | pwv < 1.9       | 2783         | 587             | 2         |
|                                                                                 | 1.9 < pwv < 3.5 | 647          | 3640            | 932       |
|                                                                                 | pwv > 3.5       | 8            | 670             | 7115      |
| Number of points 16384 ; PC=82.6%; EBD=0.1%; POD1=80.9%; POD2=74.3%; POD3=88.4% |                 |              |                 |           |

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

| precipitable water vapor<br>mm                                                  |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | pwv < 1.9    | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                                                                      | pwv < 1.9       | 3362         | 113             | 0         |
|                                                                                 | 1.9 < pwv < 3.5 | 76           | 4713            | 76        |
|                                                                                 | pwv > 3.5       | 0            | 71              | 7973      |
| Number of points 16384 ; PC=97.9%; EBD=0.0%; POD1=97.8%; POD2=96.2%; POD3=99.1% |                 |              |                 |           |

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

| total seeing                                                                   |                 | Observations |                 |           |
|--------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| arcsec                                                                         |                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| Model data                                                                     | see < 0.6       | 2421         | 1428            | 453       |
|                                                                                | 0.6 < see < 0.8 | 935          | 1201            | 1150      |
|                                                                                | see > 0.8       | 184          | 542             | 1180      |
| Number of points 9494 ; PC=50.6%; EBD=6.7%; POD1=68.4%; POD2=37.9%; POD3=42.4% |                 |              |                 |           |

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

| total seeing                                                                   |                 | Observations |                 |           |
|--------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| arcsec                                                                         |                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| Model data                                                                     | see < 0.6       | 3507         | 526             | 25        |
|                                                                                | 0.6 < see < 0.8 | 399          | 2302            | 272       |
|                                                                                | see > 0.8       | 11           | 364             | 2560      |
| Number of points 9494 ; PC=84.0%; EBD=0.4%; POD1=89.5%; POD2=72.1%; POD3=89.6% |                 |              |                 |           |

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

| total seeing                                                                   |                 | Observations |                 |           |
|--------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| arcsec                                                                         |                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| Model data                                                                     | see < 0.6       | 3390         | 0               | 316       |
|                                                                                | 0.6 < see < 0.8 | 0            | 3080            | 316       |
|                                                                                | see > 0.8       | 150          | 91              | 2467      |
| Number of points 9494 ; PC=94.1%; EBD=4.9%; POD1=95.8%; POD2=97.1%; POD3=79.6% |                 |              |                 |           |

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

| total seeing                                                                   |                 | Observations |                 |           |
|--------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| arcsec                                                                         |                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| Model data                                                                     | see < 0.6       | 3909         | 4               | 22        |
|                                                                                | 0.6 < see < 0.8 | 0            | 3172            | 21        |
|                                                                                | see > 0.8       | 8            | 16              | 2835      |
| Number of points 9494 ; PC=99.5%; EBD=0.3%; POD1=99.8%; POD2=99.4%; POD3=98.5% |                 |              |                 |           |

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

| wavefront coherence time                                                       |                    | Observations |                    |              |
|--------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                             |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                     | $\tau < 3.4$       | 1199         | 566                | 52           |
|                                                                                | $3.4 < \tau < 5.8$ | 495          | 1166               | 501          |
|                                                                                | $\tau > 5.8$       | 82           | 1063               | 3347         |
| Number of points 8471 ; PC=67.4%; EBD=1.6%; POD1=67.5%; POD2=41.7%; POD3=85.8% |                    |              |                    |              |

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

| wavefront coherence time                                                       |                    | Observations |                    |              |
|--------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                             |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                     | $\tau < 3.4$       | 1654         | 156                | 0            |
|                                                                                | $3.4 < \tau < 5.8$ | 122          | 2399               | 210          |
|                                                                                | $\tau > 5.8$       | 0            | 240                | 3690         |
| Number of points 8471 ; PC=91.4%; EBD=0.0%; POD1=93.1%; POD2=85.8%; POD3=94.6% |                    |              |                    |              |

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

| wavefront coherence time                                                       |                    | Observations |                    |              |
|--------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                             |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                     | $\tau < 3.4$       | 1563         | 41                 | 216          |
|                                                                                | $3.4 < \tau < 5.8$ | 131          | 2177               | 216          |
|                                                                                | $\tau > 5.8$       | 82           | 577                | 3684         |
| Number of points 8471 ; PC=87.6%; EBD=3.5%; POD1=88.0%; POD2=77.9%; POD3=89.5% |                    |              |                    |              |

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

| wavefront coherence time                                                       |                    | Observations |                    |              |
|--------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                             |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                     | $\tau < 3.4$       | 1773         | 1                  | 10           |
|                                                                                | $3.4 < \tau < 5.8$ | 3            | 2759               | 10           |
|                                                                                | $\tau > 5.8$       | 0            | 35                 | 3890         |
| Number of points 8471 ; PC=99.4%; EBD=0.1%; POD1=99.8%; POD2=98.7%; POD3=99.5% |                    |              |                    |              |

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

| ground layer fraction<br>dimensionless                                          | Observations    |                 |           |
|---------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                 | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                      | glf < 0.6       | 1501            | 1392      |
|                                                                                 | 0.6 < glf < 0.7 | 797             | 1137      |
|                                                                                 | glf > 0.7       | 273             | 386       |
| Number of points 8624 ; PC=38.2%; EBD=16.8%; POD1=58.4%; POD2=39.0%; POD3=21.0% |                 |                 |           |

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

| ground layer fraction<br>dimensionless                                         | Observations    |                 |           |
|--------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                     | glf < 0.6       | 2250            | 420       |
|                                                                                | 0.6 < glf < 0.7 | 312             | 2072      |
|                                                                                | glf > 0.7       | 9               | 423       |
| Number of points 8624 ; PC=80.8%; EBD=0.4%; POD1=87.5%; POD2=71.1%; POD3=84.3% |                 |                 |           |

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

| ground layer fraction<br>dimensionless                                          | Observations    |                 |           |
|---------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                 | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                      | glf < 0.6       | 2268            | 380       |
|                                                                                 | 0.6 < glf < 0.7 | 30              | 2516      |
|                                                                                 | glf > 0.7       | 273             | 19        |
| Number of points 8624 ; PC=77.2%; EBD=17.9%; POD1=88.2%; POD2=86.3%; POD3=45.1% |                 |                 |           |

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

| ground layer fraction<br>dimensionless                                         | Observations    |                 |           |
|--------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                     | glf < 0.6       | 2556            | 19        |
|                                                                                | 0.6 < glf < 0.7 | 6               | 2882      |
|                                                                                | glf > 0.7       | 9               | 14        |
| Number of points 8624 ; PC=99.1%; EBD=0.5%; POD1=99.4%; POD2=98.9%; POD3=98.0% |                 |                 |           |

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

| wind speed<br>$ms^{-1}$                                                         | Observations   |                |          |
|---------------------------------------------------------------------------------|----------------|----------------|----------|
|                                                                                 | ws < 4.2       | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 1858           | 389      |
|                                                                                 | 4.2 < ws < 7.6 | 2237           | 4554     |
|                                                                                 | ws > 7.6       | 121            | 1986     |
| Number of points 17043 ; PC=66.2%; EBD=1.0%; POD1=44.1%; POD2=65.7%; POD3=82.7% |                |                |          |

Table 23: day1 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

| wind speed<br>$ms^{-1}$                                                         | Observations   |                |          |
|---------------------------------------------------------------------------------|----------------|----------------|----------|
|                                                                                 | ws < 4.2       | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 3714           | 348      |
|                                                                                 | 4.2 < ws < 7.6 | 502            | 6196     |
|                                                                                 | ws > 7.6       | 0              | 385      |
| Number of points 17043 ; PC=90.7%; EBD=0.0%; POD1=88.1%; POD2=89.4%; POD3=94.1% |                |                |          |

Table 24: day1 - Contingency table for wind speed ( $ms^{-1}$ ) processed with AR (1H)

| wind speed<br>$ms^{-1}$                                                         | Observations     |                  |           |
|---------------------------------------------------------------------------------|------------------|------------------|-----------|
|                                                                                 | ws < 12.0        | 12.0 < ws < 18.0 | ws > 18.0 |
| Model data                                                                      | ws < 12.0        | 14605            | 166       |
|                                                                                 | 12.0 < ws < 18.0 | 131              | 1546      |
|                                                                                 | ws > 18.0        | 0                | 64        |
| Number of points 17043 ; PC=97.4%; EBD=0.0%; POD1=99.1%; POD2=87.0%; POD3=85.9% |                  |                  |           |

Table 25: day1 - Contingency table for wind speed ( $ms^{-1}$ ) processed with AR (1H) and thresholds (12.0,18.0)

## 0.2 DAY1: Contingency tables

Tables in this section are included in the range: 23-32.

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| wind direction<br>degree                                                                    |       | North | East | South | Observations<br>West |
|---------------------------------------------------------------------------------------------|-------|-------|------|-------|----------------------|
| Model data                                                                                  | North | 5178  | 360  | 16    | 99                   |
|                                                                                             | East  | 87    | 258  | 89    | 0                    |
|                                                                                             | South | 12    | 45   | 2184  | 364                  |
|                                                                                             | West  | 2015  | 16   | 269   | 3912                 |
| Number of points 14904 ; PC=77.4%; EBD=0.3%; POD1=71.0%; POD2=38.0%; POD3=85.4%; POD4=89.4% |       |       |      |       |                      |

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

| wind direction<br>degree                                                                    |       | North | East | South | Observations<br>West |
|---------------------------------------------------------------------------------------------|-------|-------|------|-------|----------------------|
| Model data                                                                                  | North | 6957  | 57   | 0     | 205                  |
|                                                                                             | East  | 49    | 605  | 9     | 0                    |
|                                                                                             | South | 0     | 17   | 2490  | 39                   |
|                                                                                             | West  | 286   | 0    | 59    | 4131                 |
| Number of points 14904 ; PC=95.2%; EBD=0.0%; POD1=95.4%; POD2=89.1%; POD3=97.3%; POD4=94.4% |       |       |      |       |                      |

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

| relative humidity<br>percent                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | rh < 7.1     | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                                                                      | rh < 7.1        | 722          | 478             | 2         |
|                                                                                 | 7.1 < rh < 15.3 | 1594         | 4903            | 750       |
|                                                                                 | rh > 15.3       | 229          | 2044            | 6299      |
| Number of points 17021 ; PC=70.1%; EBD=1.4%; POD1=28.4%; POD2=66.0%; POD3=89.3% |                 |              |                 |           |

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

| relative humidity<br>percent                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | rh < 7.1     | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                                                                      | rh < 7.1        | 2262         | 198             | 1         |
|                                                                                 | 7.1 < rh < 15.3 | 283          | 6994            | 191       |
|                                                                                 | rh > 15.3       | 0            | 233             | 6859      |
| Number of points 17021 ; PC=94.7%; EBD=0.0%; POD1=88.9%; POD2=94.2%; POD3=97.3% |                 |              |                 |           |

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

| relative humidity<br>percent                                                    |                  | Observations |                  |           |
|---------------------------------------------------------------------------------|------------------|--------------|------------------|-----------|
|                                                                                 |                  | rh < 50.0    | 50.0 < rh < 70.0 | rh > 70.0 |
| Model data                                                                      | rh < 50.0        | 16424        | 20               | 0         |
|                                                                                 | 50.0 < rh < 70.0 | 29           | 315              | 18        |
|                                                                                 | rh > 70.0        | 0            | 9                | 206       |
| Number of points 17021 ; PC=99.6%; EBD=0.0%; POD1=99.8%; POD2=91.6%; POD3=92.0% |                  |              |                  |           |

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

| precipitable water vapor<br>mm                                                  |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | pwv < 1.9    | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                                                                      | pwv < 1.9       | 1754         | 376             | 3         |
|                                                                                 | 1.9 < pwv < 3.5 | 820          | 3821            | 731       |
|                                                                                 | pwv > 3.5       | 35           | 775             | 8029      |
| Number of points 16344 ; PC=83.2%; EBD=0.2%; POD1=67.2%; POD2=76.9%; POD3=91.6% |                 |              |                 |           |

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

| precipitable water vapor<br>mm                                                  |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | pwv < 1.9    | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                                                                      | pwv < 1.9       | 2505         | 88              | 0         |
|                                                                                 | 1.9 < pwv < 3.5 | 104          | 4808            | 105       |
|                                                                                 | pwv > 3.5       | 0            | 76              | 8658      |
| Number of points 16344 ; PC=97.7%; EBD=0.0%; POD1=96.0%; POD2=96.7%; POD3=98.8% |                 |              |                 |           |

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

| wind speed<br>$ms^{-1}$                                                         | Observations   |                |          |
|---------------------------------------------------------------------------------|----------------|----------------|----------|
|                                                                                 | ws < 4.2       | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 3531           | 1311     |
|                                                                                 | 4.2 < ws < 7.6 | 2233           | 2651     |
|                                                                                 | ws > 7.6       | 252            | 1221     |
| Number of points 17072 ; PC=64.2%; EBD=2.0%; POD1=58.7%; POD2=51.1%; POD3=81.4% |                |                |          |

Table 33: night2 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

0.3 NIGHT2: Contingency tables

Tables in this section are included in the range: 33-42.

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| wind direction<br>degree | Observations |      |       |      |
|--------------------------|--------------|------|-------|------|
|                          | North        | East | South | West |
| Model data               | North        | 6647 | 480   | 66   |
|                          | East         | 547  | 777   | 632  |
|                          | South        | 41   | 268   | 2898 |
|                          | West         | 103  | 79    | 82   |

Number of points 13177 ; PC=79.6%; EBD=1.5%; POD1=90.6%; POD2=48.4%; POD3=78.8%; POD4=29.6%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent | Observations    |                 |           |
|------------------------------|-----------------|-----------------|-----------|
|                              | rh < 7.1        | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                   | rh < 7.1        | 1888            | 696       |
|                              | 7.1 < rh < 15.3 | 2482            | 3533      |
|                              | rh > 15.3       | 642             | 1948      |

Number of points 16950 ; PC=59.5%; EBD=4.1%; POD1=37.7%; POD2=57.2%; POD3=80.9%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm | Observations    |                 |           |
|--------------------------------|-----------------|-----------------|-----------|
|                                | pwv < 1.9       | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                     | pwv < 1.9       | 2805            | 715       |
|                                | 1.9 < pwv < 3.5 | 669             | 3133      |
|                                | pwv > 3.5       | 19              | 883       |

Number of points 16168 ; PC=80.1%; EBD=0.4%; POD1=80.3%; POD2=66.2%; POD3=88.3%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

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| total seeing<br>arcsec                                       |                 | Observations |                 |           |
|--------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                              |                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| Model data                                                   | see < 0.6       | 2662         | 1472            | 581       |
|                                                              | 0.6 < see < 0.8 | 1118         | 1410            | 1232      |
|                                                              | see > 0.8       | 290          | 855             | 1603      |
| Number of points 11223 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=% |                 |              |                 |           |

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

| total seeing<br>arcsec                                       |                 | Observations |                 |           |
|--------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                              |                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| Model data                                                   | see < 0.6       | 3855         | 0               | 397       |
|                                                              | 0.6 < see < 0.8 | 0            | 3557            | 397       |
|                                                              | see > 0.8       | 215          | 180             | 3019      |
| Number of points 11223 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=% |                 |              |                 |           |

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

| wavefront coherence time<br>ms                                                  |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | tau < 3.4    | 3.4 < tau < 5.8 | tau > 5.8 |
| Model data                                                                      | tau < 3.4       | 1525         | 581             | 61        |
|                                                                                 | 3.4 < tau < 5.8 | 578          | 1344            | 528       |
|                                                                                 | tau > 5.8       | 97           | 1325            | 4054      |
| Number of points 10093 ; PC=68.6%; EBD=1.6%; POD1=69.3%; POD2=41.4%; POD3=87.3% |                 |              |                 |           |

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

| wavefront coherence time<br>ms                                                  |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | tau < 3.4    | 3.4 < tau < 5.8 | tau > 5.8 |
| Model data                                                                      | tau < 3.4       | 1939         | 49              | 268       |
|                                                                                 | 3.4 < tau < 5.8 | 164          | 2282            | 268       |
|                                                                                 | tau > 5.8       | 97           | 919             | 4375      |
| Number of points 10093 ; PC=85.2%; EBD=3.6%; POD1=88.1%; POD2=70.2%; POD3=89.1% |                 |              |                 |           |

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

| ground layer fraction<br>dimensionless                                           | Observations    |                 |           |
|----------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                  | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 1557            | 1643      |
|                                                                                  | 0.6 < glf < 0.7 | 1063            | 1518      |
|                                                                                  | glf > 0.7       | 250             | 375       |
| Number of points 10289 ; PC=38.1%; EBD=15.6%; POD1=54.3%; POD2=42.9%; POD3=21.7% |                 |                 |           |

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

| ground layer fraction<br>dimensionless                                           | Observations    |                 |           |
|----------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                  | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 2585            | 384       |
|                                                                                  | 0.6 < glf < 0.7 | 35              | 3128      |
|                                                                                  | glf > 0.7       | 250             | 24        |
| Number of points 10289 ; PC=78.8%; EBD=16.9%; POD1=90.1%; POD2=88.5%; POD3=46.9% |                 |                 |           |

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

| wind speed<br>$ms^{-1}$                                                         |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
|                                                                                 |                | ws < 4.2     | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 1910         | 413            | 69       |
|                                                                                 | 4.2 < ws < 7.6 | 2292         | 4535           | 1181     |
|                                                                                 | ws > 7.6       | 94           | 1933           | 4753     |
| Number of points 17180 ; PC=65.2%; EBD=0.9%; POD1=44.5%; POD2=65.9%; POD3=79.2% |                |              |                |          |

Table 43: day2 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

0.4 DAY2: Contingency tables

Tables in this section are included in the range: 43-46.

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| wind direction<br>degree | Observations |      |       |      |
|--------------------------|--------------|------|-------|------|
|                          | North        | East | South | West |
| Model data               | North        | 5074 | 368   | 19   |
|                          | East         | 152  | 210   | 86   |
|                          | South        | 16   | 36    | 2012 |
|                          | West         | 2094 | 40    | 478  |

Number of points 14984 ; PC=74.6%; EBD=0.5%; POD1=69.2%; POD2=32.1%; POD3=77.5%; POD4=88.2%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent | Observations    |                 |           |
|------------------------------|-----------------|-----------------|-----------|
|                              | rh < 7.1        | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                   | rh < 7.1        | 582             | 413       |
|                              | 7.1 < rh < 15.3 | 1732            | 4329      |
|                              | rh > 15.3       | 344             | 2739      |

Number of points 17155 ; PC=64.6%; EBD=2.0%; POD1=21.9%; POD2=57.9%; POD3=88.1%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm | Observations    |                 |           |
|--------------------------------|-----------------|-----------------|-----------|
|                                | pwv < 1.9       | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                     | pwv < 1.9       | 1818            | 552       |
|                                | 1.9 < pwv < 3.5 | 963             | 3548      |
|                                | pwv > 3.5       | 37              | 989       |

Number of points 16510 ; PC=79.0%; EBD=0.3%; POD1=64.5%; POD2=69.7%; POD3=89.3%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

| wind speed<br>$ms^{-1}$                                                         | Observations   |                |          |
|---------------------------------------------------------------------------------|----------------|----------------|----------|
|                                                                                 | ws < 4.2       | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 3483           | 1411     |
|                                                                                 | 4.2 < ws < 7.6 | 2232           | 2540     |
|                                                                                 | ws > 7.6       | 327            | 1174     |
| Number of points 17005 ; PC=62.2%; EBD=2.9%; POD1=57.6%; POD2=49.6%; POD3=78.1% |                |                |          |

Table 47: night3 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

0.5 NIGHT3: Contingency tables

Tables in this section are included in the range: 47-56.

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| wind direction<br>degree | Observations |      |       |      |
|--------------------------|--------------|------|-------|------|
|                          | North        | East | South | West |
| Model data               | North        | 6484 | 562   | 81   |
|                          | East         | 548  | 745   | 640  |
|                          | South        | 57   | 247   | 2891 |
|                          | West         | 115  | 50    | 104  |

Number of points 13081 ; PC=78.7%; EBD=1.5%; POD1=90.0%; POD2=46.4%; POD3=77.8%; POD4=32.1%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent | Observations    |                 |           |
|------------------------------|-----------------|-----------------|-----------|
|                              | rh < 7.1        | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                   | rh < 7.1        | 1689            | 786       |
|                              | 7.1 < rh < 15.3 | 2577            | 3502      |
|                              | rh > 15.3       | 712             | 1915      |

Number of points 16883 ; PC=56.7%; EBD=4.5%; POD1=33.9%; POD2=56.5%; POD3=76.9%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm | Observations    |                 |           |
|--------------------------------|-----------------|-----------------|-----------|
|                                | pwv < 1.9       | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                     | pwv < 1.9       | 2765            | 900       |
|                                | 1.9 < pwv < 3.5 | 673             | 2794      |
|                                | pwv > 3.5       | 57              | 1034      |

Number of points 16101 ; PC=76.0%; EBD=0.9%; POD1=79.1%; POD2=59.1%; POD3=84.8%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

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| total seeing                                                 |                 | Observations |                 |           |
|--------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| arcsec                                                       |                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| Model data                                                   | see < 0.6       | 2649         | 1562            | 576       |
|                                                              | 0.6 < see < 0.8 | 1181         | 1333            | 1291      |
|                                                              | see > 0.8       | 243          | 832             | 1509      |
| Number of points 11176 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=% |                 |              |                 |           |

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

| total seeing                                                 |                 | Observations |                 |           |
|--------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| arcsec                                                       |                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| Model data                                                   | see < 0.6       | 3868         | 0               | 407       |
|                                                              | 0.6 < see < 0.8 | 0            | 3596            | 407       |
|                                                              | see > 0.8       | 205          | 131             | 2969      |
| Number of points 11176 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=% |                 |              |                 |           |

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

| wavefront coherence time                                                        |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| ms                                                                              |                 | tau < 3.4    | 3.4 < tau < 5.8 | tau > 5.8 |
| Model data                                                                      | tau < 3.4       | 1531         | 582             | 44        |
|                                                                                 | 3.4 < tau < 5.8 | 567          | 1287            | 603       |
|                                                                                 | tau > 5.8       | 103          | 1378            | 4000      |
| Number of points 10095 ; PC=67.5%; EBD=1.5%; POD1=69.6%; POD2=39.6%; POD3=86.1% |                 |              |                 |           |

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

| wavefront coherence time                                                        |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
| ms                                                                              |                 | tau < 3.4    | 3.4 < tau < 5.8 | tau > 5.8 |
| Model data                                                                      | tau < 3.4       | 1916         | 45              | 283       |
|                                                                                 | 3.4 < tau < 5.8 | 182          | 2327            | 283       |
|                                                                                 | tau > 5.8       | 103          | 875             | 4364      |
| Number of points 10095 ; PC=85.3%; EBD=3.8%; POD1=87.1%; POD2=71.7%; POD3=88.5% |                 |              |                 |           |

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

| ground layer fraction<br>dimensionless                                           | Observations    |                 |           |
|----------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                  | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 1476            | 1522      |
|                                                                                  | 0.6 < glf < 0.7 | 1149            | 1623      |
|                                                                                  | glf > 0.7       | 251             | 384       |
| Number of points 10291 ; PC=37.6%; EBD=16.2%; POD1=51.3%; POD2=46.0%; POD3=19.8% |                 |                 |           |

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

| ground layer fraction<br>dimensionless                                           | Observations    |                 |           |
|----------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                  | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 2571            | 328       |
|                                                                                  | 0.6 < glf < 0.7 | 54              | 3176      |
|                                                                                  | glf > 0.7       | 251             | 25        |
| Number of points 10291 ; PC=78.8%; EBD=17.3%; POD1=89.4%; POD2=90.0%; POD3=45.5% |                 |                 |           |

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

| wind speed<br>$ms^{-1}$                                                         | Observations     |                  |            |
|---------------------------------------------------------------------------------|------------------|------------------|------------|
|                                                                                 | $ws < 4.2$       | $4.2 < ws < 7.6$ | $ws > 7.6$ |
| Model data                                                                      | $ws < 4.2$       | 1886             | 498        |
|                                                                                 | $4.2 < ws < 7.6$ | 2220             | 4529       |
|                                                                                 | $ws > 7.6$       | 162              | 1862       |
| Number of points 17113 ; PC=64.9%; EBD=1.4%; POD1=44.2%; POD2=65.7%; POD3=78.6% |                  |                  |            |

Table 57: day3 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

0.6 DAY3: Contingency tables

Tables in this section are included in the range: 57-60.

| wind direction<br>degree | Observations |      |       |      |
|--------------------------|--------------|------|-------|------|
|                          | North        | East | South | West |
| Model data               | North        | 4922 | 411   | 41   |
|                          | East         | 126  | 178   | 95   |
|                          | South        | 13   | 30    | 1963 |
|                          | West         | 2269 | 41    | 496  |

Number of points 14939 ; PC=72.2%; EBD=0.6%; POD1=67.1%; POD2=27.0%; POD3=75.6%; POD4=85.6%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent | Observations    |                 |           |
|------------------------------|-----------------|-----------------|-----------|
|                              | rh < 7.1        | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                   | rh < 7.1        | 393             | 458       |
|                              | 7.1 < rh < 15.3 | 1910            | 4252      |
|                              | rh > 15.3       | 338             | 2769      |

Number of points 17088 ; PC=62.5%; EBD=2.1%; POD1=14.9%; POD2=56.9%; POD3=86.7%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm | Observations    |                 |           |
|--------------------------------|-----------------|-----------------|-----------|
|                                | pwv < 1.9       | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                     | pwv < 1.9       | 1745            | 513       |
|                                | 1.9 < pwv < 3.5 | 1078            | 3499      |
|                                | pwv > 3.5       | 34              | 951       |

Number of points 16490 ; PC=76.8%; EBD=0.6%; POD1=61.1%; POD2=70.5%; POD3=85.6%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration